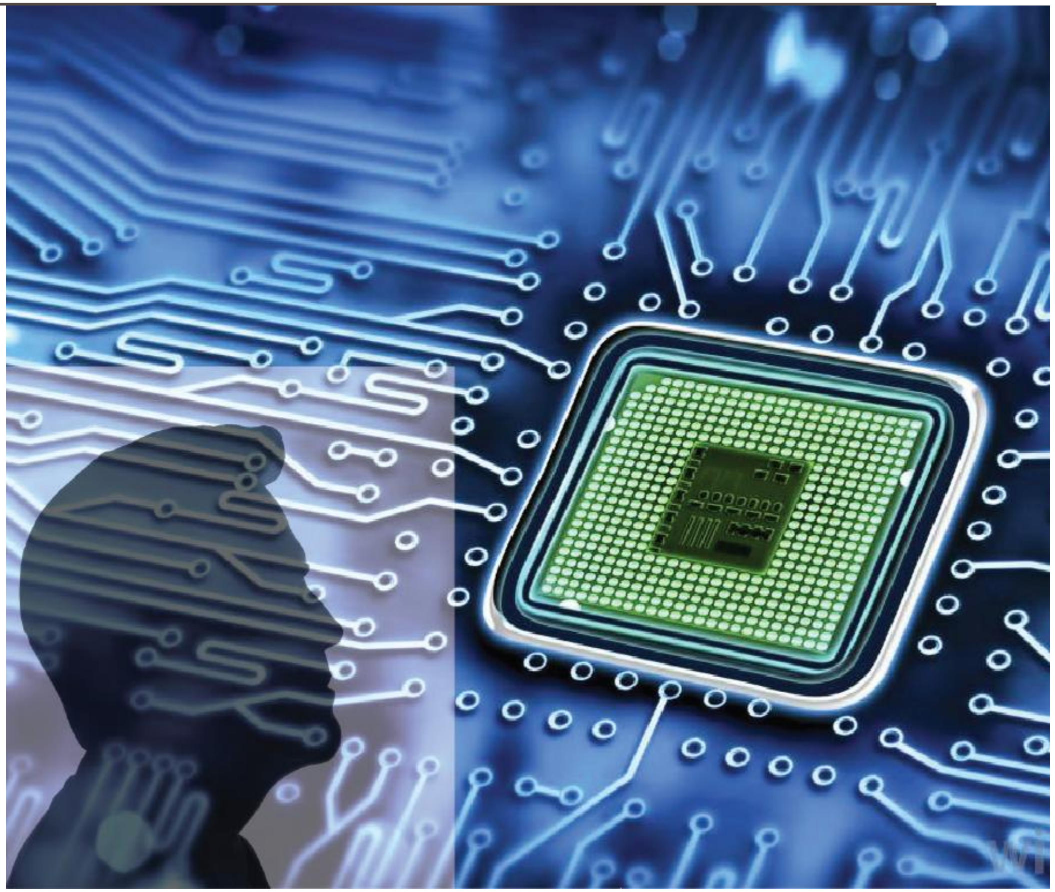
# ADVANCED DIGITAL DESIGN AND VERIFICATION

## Week-14 Lecture-1
## Clock Domain Crossing (CDC)



Sneh Saurabh
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# Clock Domain Crossing (CDC)

# Clock Domains and Clock Domain Crossing (CDC)

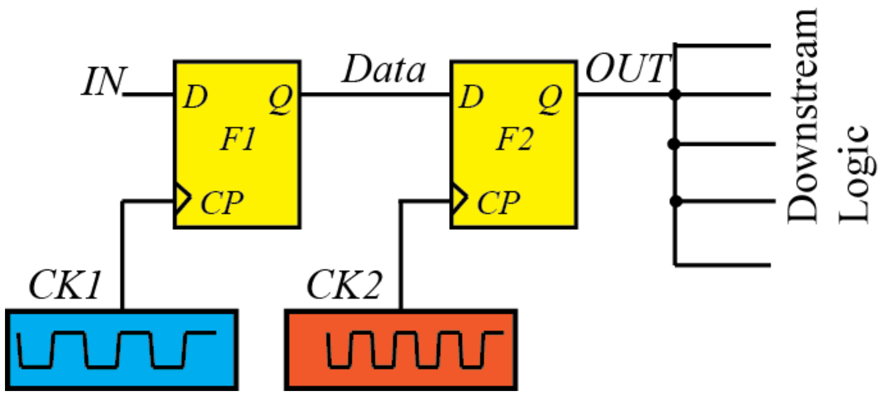
- Multiple clock frequencies are used in SoCs
  - Optimize power
  - Availability of IPs

**Clock Domain**

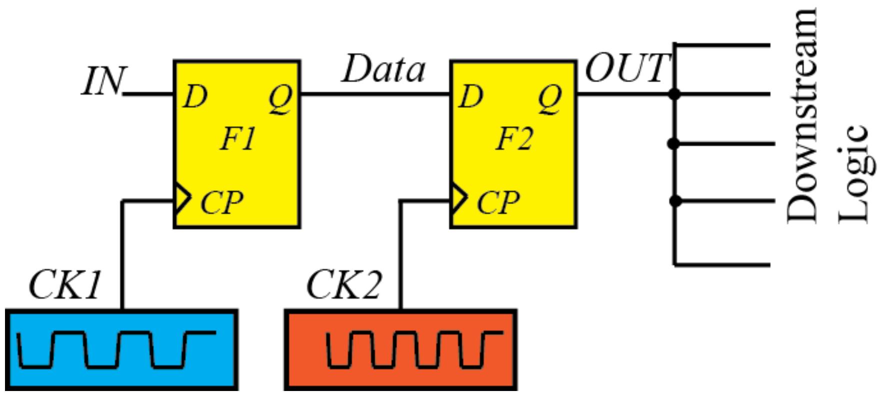
- Flip-flops that are clocked by the same clock source are said to be in the same clock domain
- Flip-flops of the same clock domain operate at the same clock frequency and have a fixed phase relationship

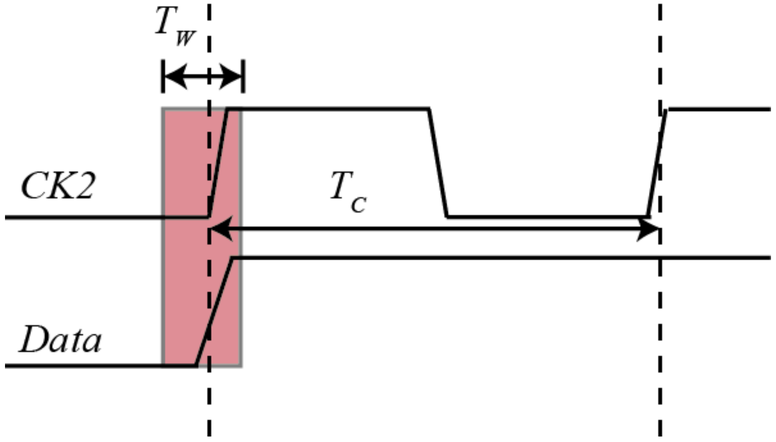
**Clock Domain Crossing (CDC)**

- When data is launched in one clock domain and captured in another clock domain, CDC is said to occur



# Forbidden Event in a Flip-flop




## Clock Domain Crossing (CDC)

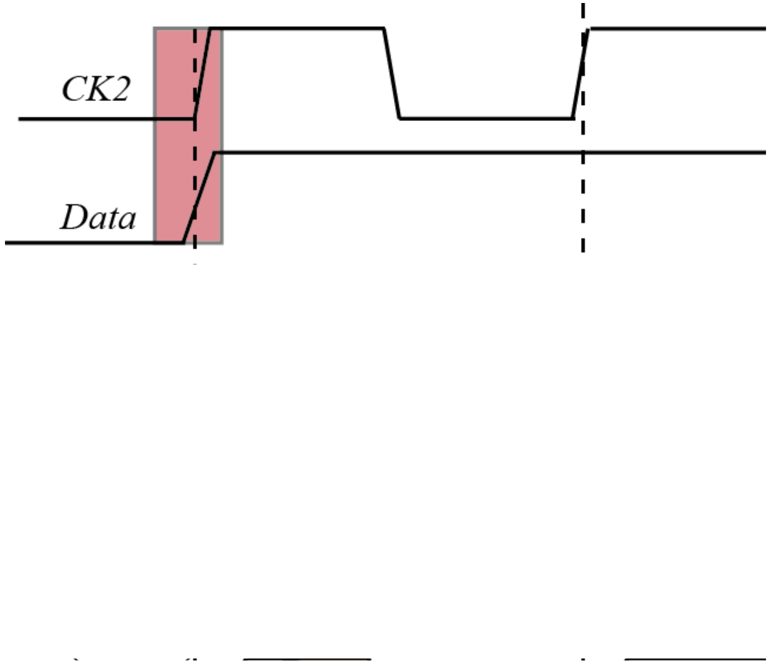
- Clocks are asynchronous
- *Data* can change very close to the clock edge of *CK2*
- *Data* change occurs within the forbidden window defined by the setup-hold time of *F2*
    - Forbidden event: setup/hold time violations occur for *F2*

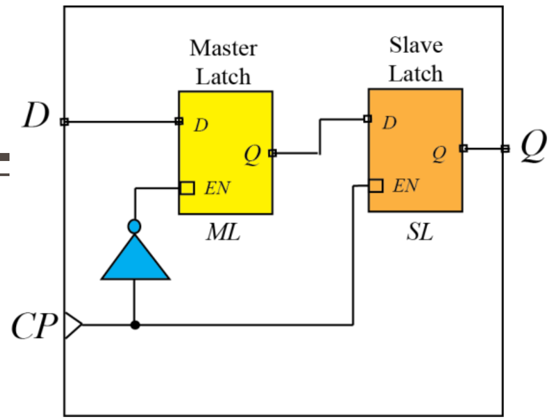
## Frequency of forbidden event

- Assume Data change occurs uniformly in the clock period
- Assume frequency of *Data* is $f_D$
- Frequency of forbidden event $f_{FE}$ is:

$$f_{FE} = \frac{T_W}{T_C} \times f_D$$

# Consequences of Forbidden Event




**Case 1**

- ML latches the data properly or it enters into a metastable state and quickly resolves to stable 1
- SL latches the data correctly and produces a stable 1

**Case 2**

- ML misses the data completely or it enters into a metastable state and quickly resolves to 0.
- SL latches the data correctly in the next clock cycle

**Case 3**

- ML goes into a metastable state and resolves to 1 within the first clock cycle
- SL produces a metastable output initially, but it resolves to 1 within the first cycle
- Effective clock to Q delay is significantly high

**Case 4**

- ML goes into a metastable state. SL produces a 1 (pull-up)
- Then, ML resolves to 0 and SL follows it by toggling its output value.
- In the next cycle, both ML and SL produce the data correctly.

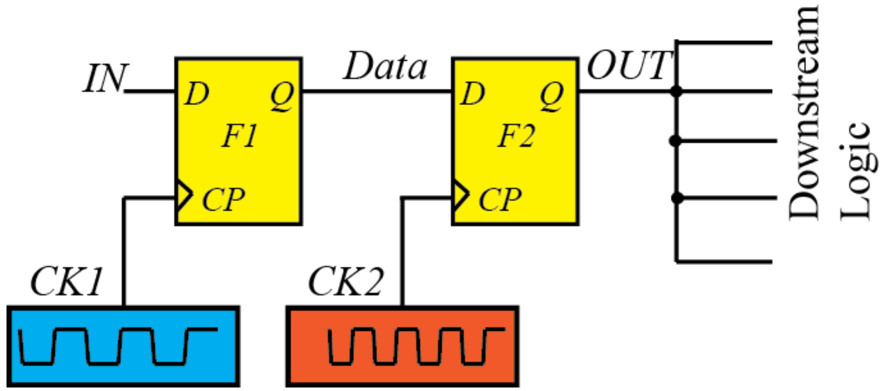
**Summary**

- Behavior unpredictable (cycle of data capture and delay)

# Problems associated with CDC

**Problems**

1. The downstream logic elements can receive and interpret contrasting logic levels for the same metastable output.
   - Can lead to a functional problem in the circuit.
2. Due to the metastable voltage at the gate terminal, the pull-up/pull-down transistors in the downstream logic elements can draw a large current.
3. Due to an unexpected large CP→Q delay in F2, there can be timing violations in the downstream logic elements.




**Solution**

- Need to mask the metastable voltage of *F2* from the downstream logic
- Add synchronizers

# Theory of Synchronizers

- A metastable latch resolves to 0/1 after some time
- If a latch is metastable at time $t = 0$ and it remains metastable after waiting for $t = S$, we say that a failure has occurred
- The probability of the occurrence of failure is $e^{-S/\tau}$, where $\tau$ depends on the device parameters and the fabrication technology

### Failure Rate

- Frequency of forbidden event $f_D$ is:

$$f_{FE} = \frac{T_W}{T_C} \times f_D$$

- Failure rate is:

$$FR = \frac{T_W}{T_C} \times f_D \times e^{-S/\tau}$$

- Probability of failure decreases exponentially with time
  - Implies that if we wait long enough, the output will settle to 0/1 with a high probability
  - This observation forms the basis of designing special circuits known as synchronizer

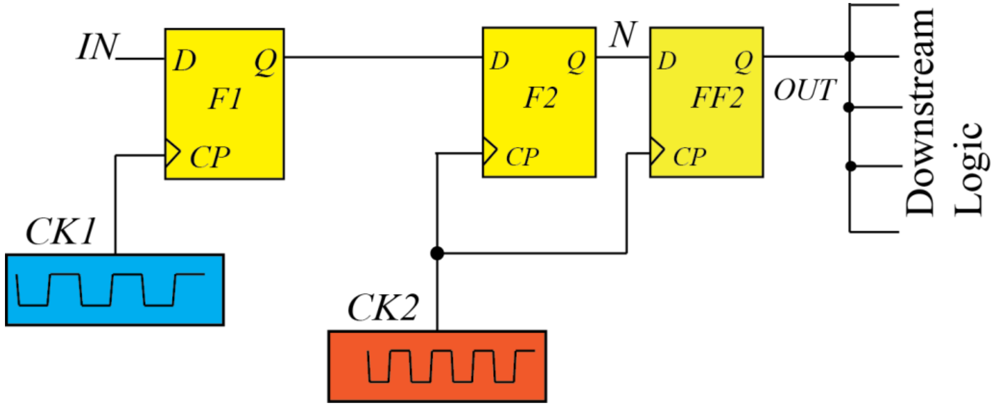
### Mean Time Between Failure (MTBF)

- Mean time between failure is the reciprocal of failure rate:

$$MTBF = \frac{T_C}{T_W \times f_D} \times e^{S/\tau}$$

- **In practice**: make MTBF several orders of time larger than the expected product lifetime using synchronizers

# Double Flip-flop Synchronizer (1)

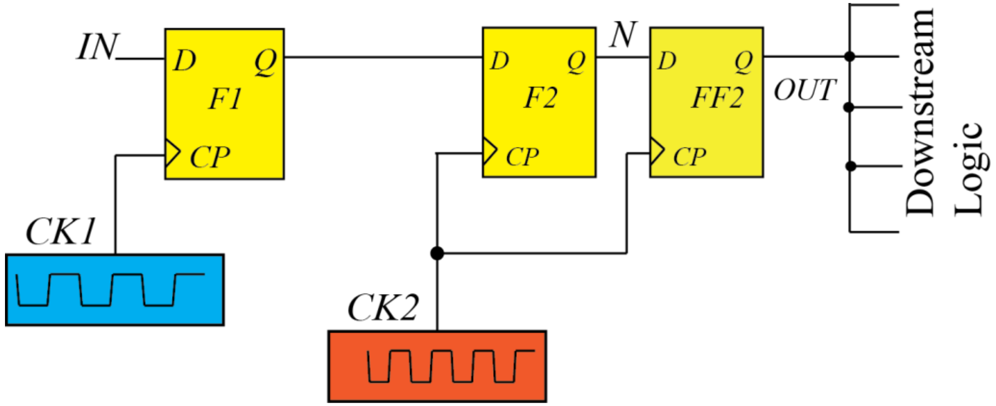



- Capture flip-flop *F2* can go into a metastable state
- Capture the output of *F2* in the next clock cycle using *FF2*

- Downstream logic uses the output of *FF2* and not *F2*
- Waiting time for *FF2* before sampling:
  $S = T_C - T_{C \to Q,F2} - T_{setup,FF2} - T_{wire,N}$
- By keeping $S$ high enough, acceptable MTBF can be obtained

- *FF2* will have a stable 0/1 (and not a metastable voltage level), except for once every MTBF years
  - ➢ Correct Data will be available at the output of *FF2* after second clock cycle (cases 1 and 3) or third clock cycle (cases 2 and 4)

# Double Flip-flop Synchronizer (2)



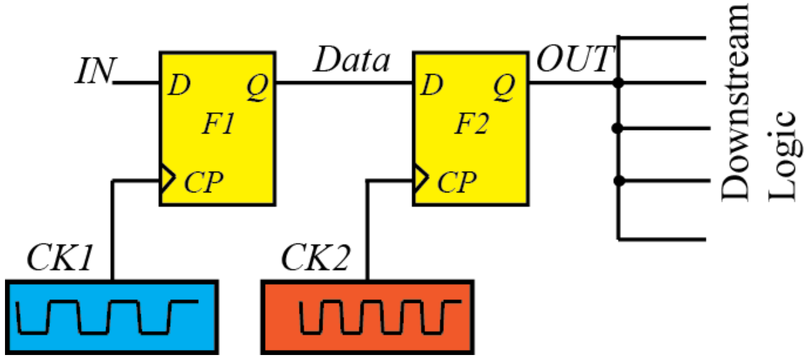

- There is still uncertainty of one clock cycle in the clock domain CK2.
  - Can handle this functionally by ensuring that the downstream logic of *FF2* works correctly when *FF2* provides *Data* in either the second or the third clock cycle.

- Primary purpose of the synchronizer is to shield the downstream logic from the metastable voltage levels with a high probability.
  - It is achieved by the double flip-flop synchronizer satisfactorily in most cases

Demerits:

- Incurs a penalty of one clock cycle in the latency for data propagation
- Area increase

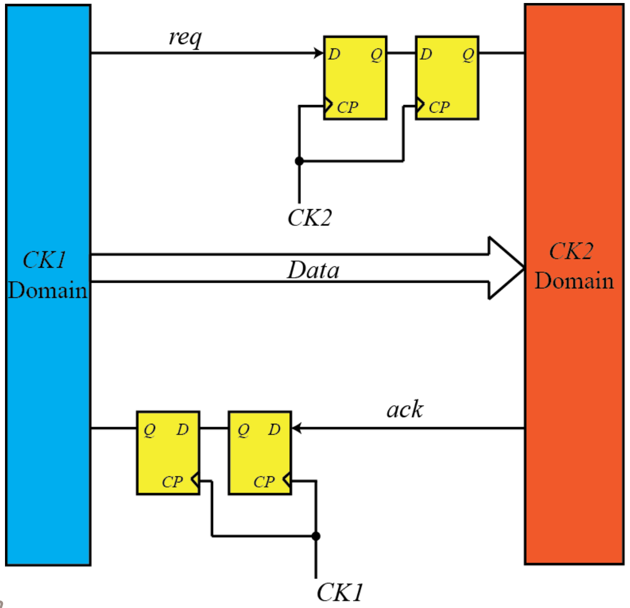
# Ensuring Input Data is Stable



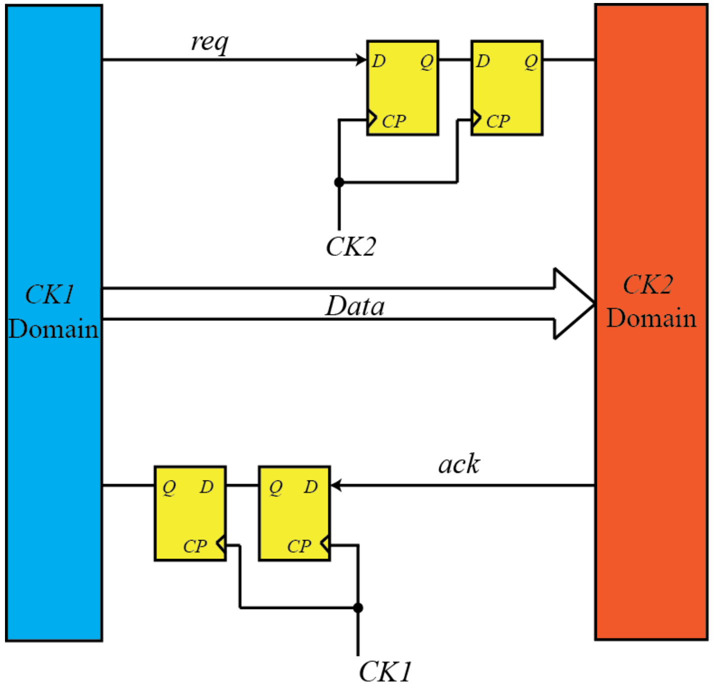

- Assumption: If *Data* is missed in the first clock cycle or the metastable value settles to the previous state, it is expected to get captured in the next clock cycle.
  - ➢ Is guaranteed only when *Data* is held constant for at least two clock cycles at *F2*.

To ensure this:

- Feedback mechanism from the receiver domain *CK2* to the sender domain *CK1*
- Request-acknowledge protocol between the clock domains
- The sender does not change *Data* unless it receives an acknowledge signal from the receiver.



# Data Coherency Challenges




- For sending multi-bit data, synchronizing them bitwise will cause problem:
  - ➢ Can lead to some bits being updated one cycle before others
  - ➢ Received multi-bit data can be incoherent and can cause functional problems
- **Solution:** Synchronize the control signal rather than the multi-bit data signal

- Should not synchronize a signal from another clock domain using multiple synchronizers in parallel.
  - ➢ The same signal can be interpreted as both 0 and 1 in the downstream logic and can lead to functional problems.

# CDC: Verification

- CDC poses nontrivial verification challenges
- Cannot verify the synchronization of asynchronous clock domains using STA

- We use a combination of :
  - Topological analysis: to detect CDC
  - Pattern matching: to check for existence of some synchronizers
  - Formal techniques: adherence to a given CDC protocol

# Reset Domain Crossing (RDC)

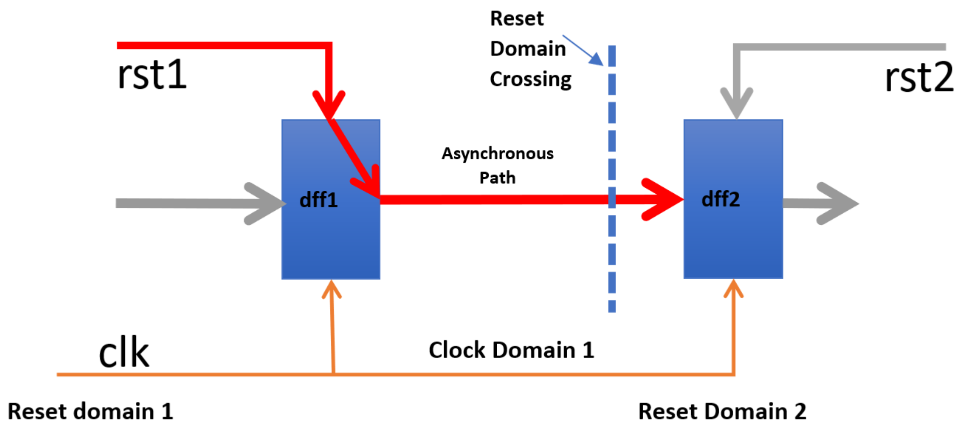
# Reset Domain Crossing: Basics

## Reset Domain

- A collection of flip-flops that are all controlled by the same reset signal.
- Reset signal is asserted: every flop in that domain is forced to a known value.

## Reset Domain Crossing

- When a signal crosses from one reset domain into another
- Receiving flip-flop (in Reset Domain 2) can sample a signal in the forbidden window
  - ➢ Can go into metastable state





- When a signal crosses from one reset domain into another
- Receiving flip-flop (in Reset Domain 2) can sample a signal in the forbidden window
  - ➢ Can go into metastable state

# References


- S. Saurabh, "Introduction to VLSI Design Flow". Cambridge: *Cambridge University Press*, 2023.